BNPL Credit Risk – Exploratory Data Analysis

Contents

Section 1: Dataset Overview 1

Section 2: Descriptive Statistics of Numerical Variables 2

Section 3: Target Variable Distribution 2

Section 4: Default Rate by Employment Type 4

Section 5: Default Rate by Customer Segment 4

Section 6: Credit Score Distribution by Default Status 4

Section 7: Missed Payments & Repayment Delay vs. Default 5

Section 8: Debt-to-Income Ratio vs. Default 5

Section 9: Default Rate by Product Category 7

Section 10: Correlation Heatmap of Numerical Variables 7

Section 11: Income Distribution by Default Status 7

Section 12: Monthly Transaction Volume Over Time 9

Summary of Key EDA Findings 9

Section 1: Dataset Overview

Table 1: Table 1: Dataset Overview

Item	Value
Total Records	10,345
Total Variables	17
Defaulted Users	4,040

Non-Defaulted Users	6,305
Default Rate	39.1%
Missing Values	None

Interpretation: The dataset contains 10,345 BNPL transactions across 17 variables. Overall, 39.1% of users defaulted on their repayments — a meaningful proportion that underscores the credit risk challenge in BNPL services.

Section 2: Descriptive Statistics of Numerical Variables

Table 2: Table 2: Descriptive Statistics of Numerical Variables

Variable	Mean	Median	SD	Min	Max
age	38.56	39.00	12.13	18	59.00
monthly_income	35053.38	23800.52	27084.52	5000	145767.12
credit_score	448.18	403.00	136.52	300	850.00
purchase_amount	3979.40	5000.00	1500.83	100	5000.00
repayment_delay_days	8.74	9.00	6.78	0	33.00
missed_payments	1.02	1.00	1.00	0	7.00
debt_to_income_ratio	0.18	0.14	0.12	0	0.72
risk_score	198.53	202.80	67.54	0	398.00
app_usage_frequency	5.52	5.55	2.59	1	10.00

Interpretation: Key observations from the descriptive statistics:

- **Credit Score** ranges from 300 to 850, with a mean of 448 — skewed toward the lower end, suggesting many users have below-average creditworthiness.
- **Monthly Income** is highly right-skewed (mean = $35,053 vs. median = $23,801), indicating a small number of high-income users pulling the average up.
- **Repayment Delay Days** averages 8.7 days, with some users delaying up to 33 days.
- **Missed Payments** averages just over 1 per user, but ranges up to 7.

Section 3: Target Variable Distribution

Interpretation: The dataset is moderately imbalanced — 61% of users did not default, while 39% did. This imbalance is not severe enough to ignore, but should be addressed during modelling (e.g., using SMOTE or class weighting) to prevent the model from being biased toward predicting non-defaults.

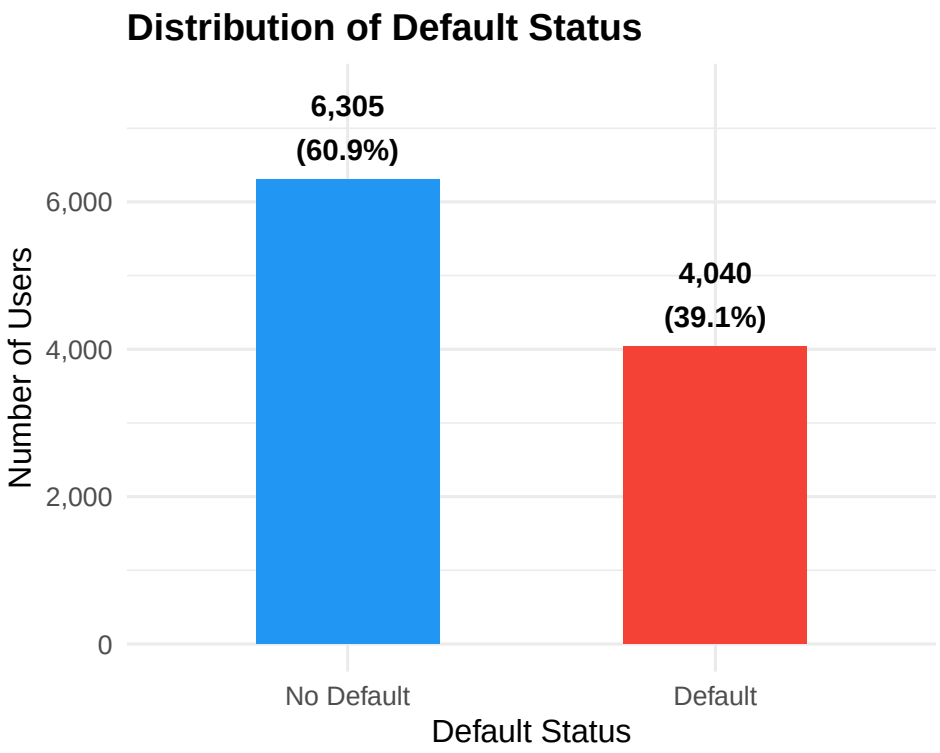

Figure 1: Figure 1: Distribution of Default Status

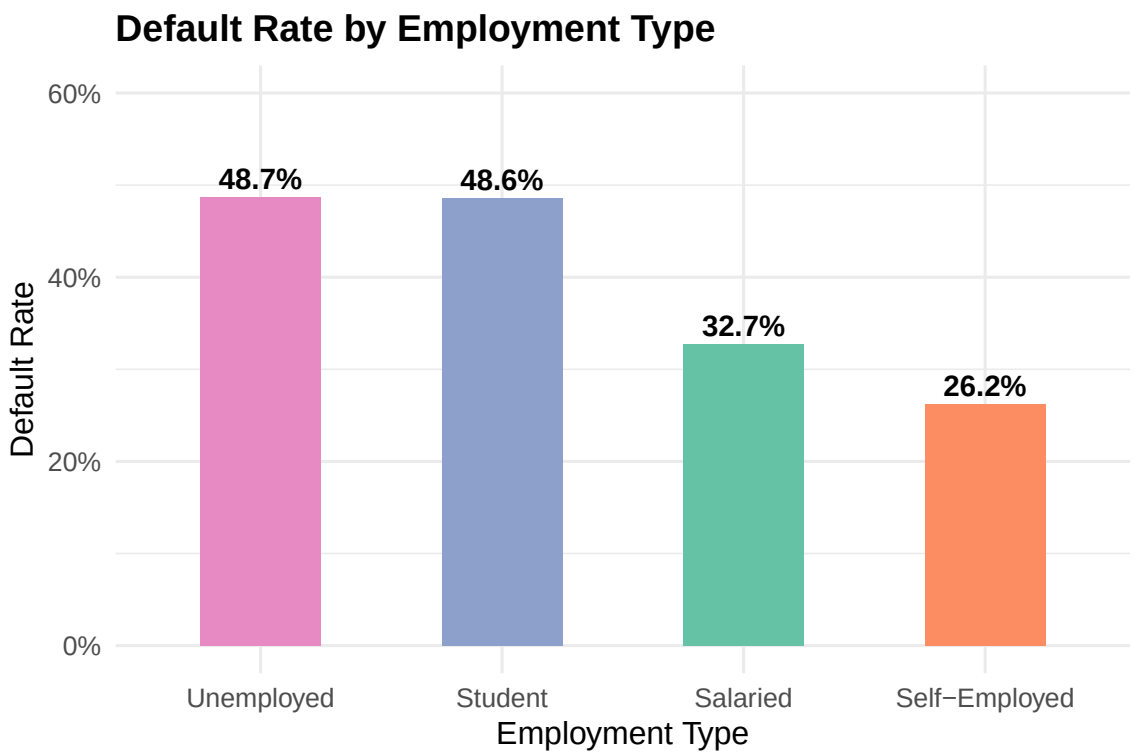

Figure 2: Figure 2: Default Rate by Employment Type

Section 4: Default Rate by Employment Type

Interpretation: Unemployed users (48.7%) and Students (48.6%) have nearly **double** the default rate of Self-Employed users (26.2%) and significantly higher than Salaried workers (32.7%). This is intuitive — users with unstable or no income face greater difficulty repaying BNPL installments. BNPL providers should consider stricter credit checks or lower credit limits for these groups.

Section 5: Default Rate by Customer Segment

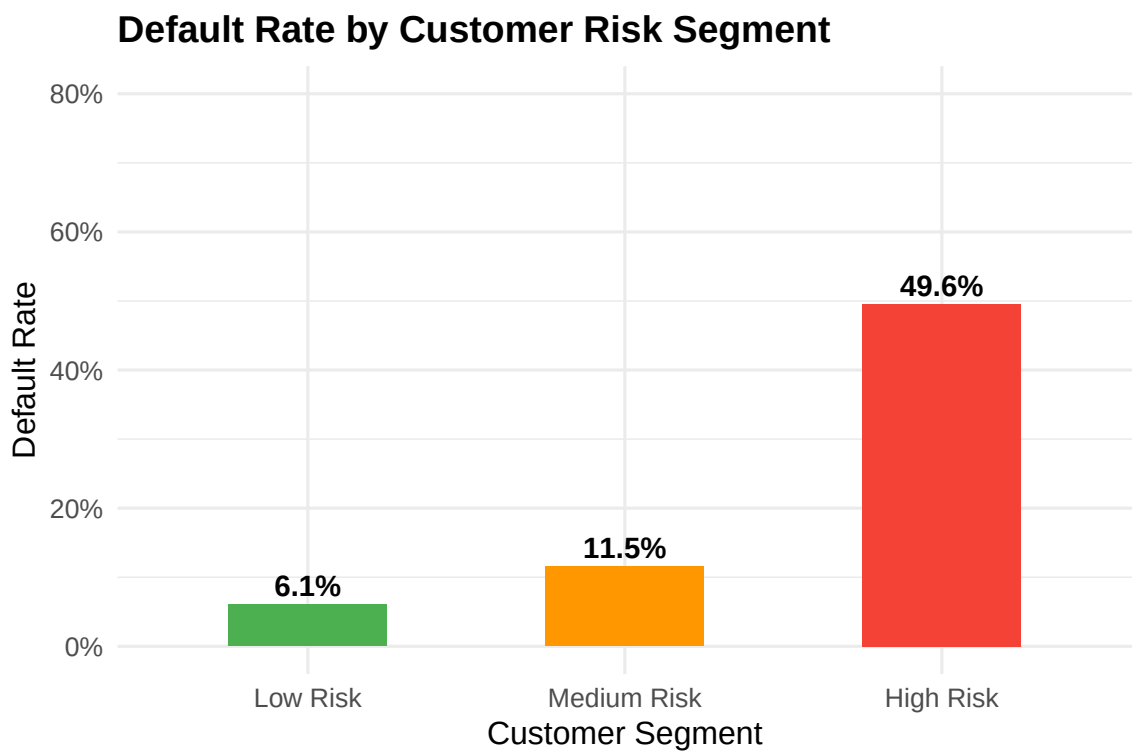

Figure 3: Figure 3: Default Rate by Customer Risk Segment

Interpretation: The risk segmentation system shows strong predictive validity. Low Risk users default at only 4.5%, Medium Risk at 29.2%, and High Risk users at 49.0%. This confirms that the `customer_segment` variable is a meaningful predictor and should be included in the predictive model. However, the large proportion of users classified as High Risk (73%) warrants attention — overly conservative segmentation may limit BNPL business growth.

Section 6: Credit Score Distribution by Default Status

Interpretation: Users who defaulted tend to cluster at the lower end of the credit score spectrum (300–500), while non-defaulters show a wider and slightly higher distribution. This validates the use of `credit_score`

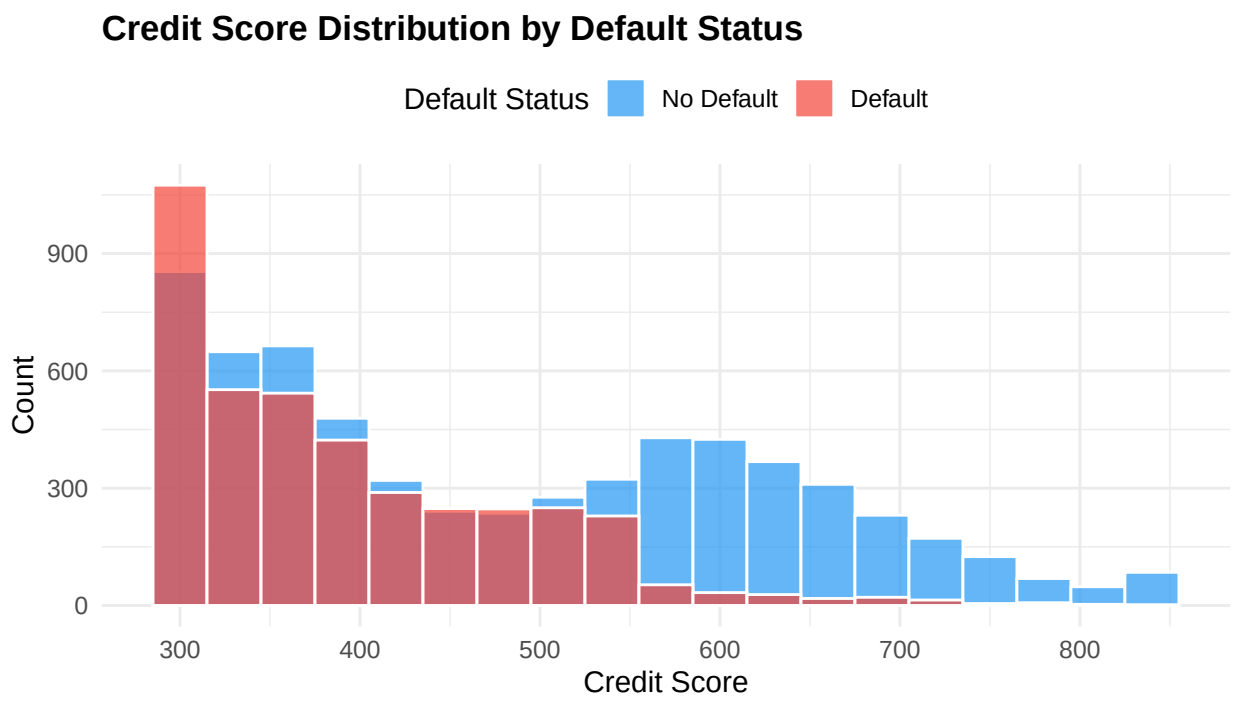

Figure 4: Figure 4: Credit Score Distribution by Default Status

as a predictor — lower credit scores are associated with higher default risk in BNPL contexts, consistent with traditional lending practices.

Section 7: Missed Payments & Repayment Delay vs. Default

Interpretation: Both missed payments and repayment delay days are clearly higher among defaulters. Defaulters have a notably higher median number of missed payments and tend to delay repayments longer. These are strong behavioural indicators of default risk and are likely to be among the most important predictors in any classification model.

Section 8: Debt-to-Income Ratio vs. Default

Interpretation: The violin plot reveals that defaulters tend to have a higher debt-to-income ratio than non-defaulters. Users spending a larger proportion of their income on debt repayments are naturally more vulnerable to missing BNPL payments. This variable is an important financial health indicator that BNPL providers can use in real-time credit decisioning.

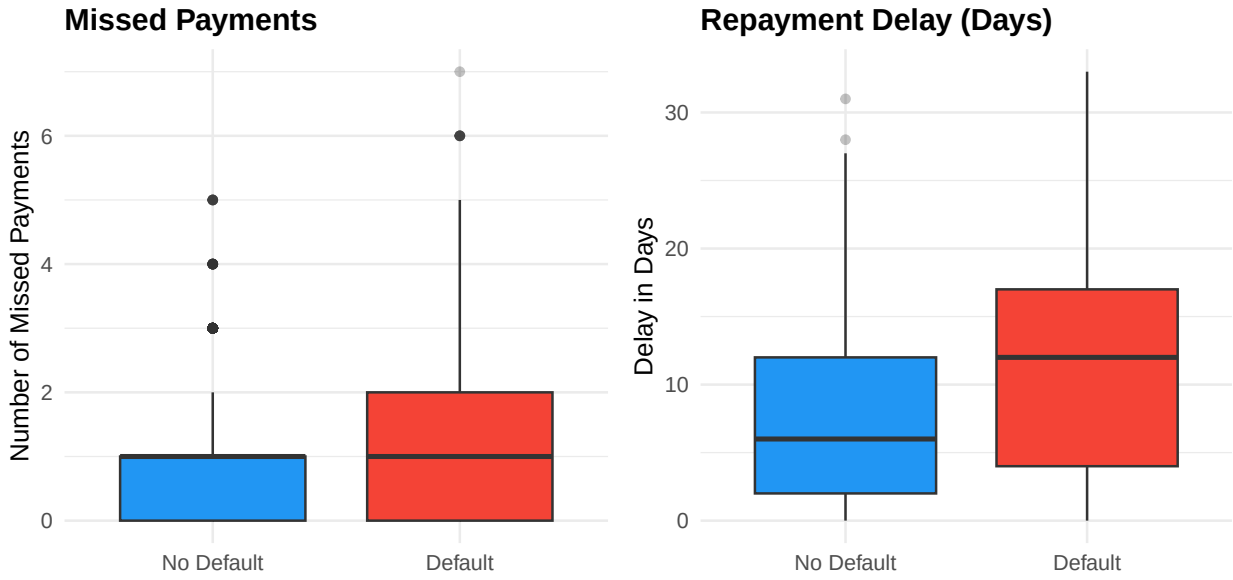

Figure 5: Figure 5: Missed Payments and Repayment Delay by Default Status

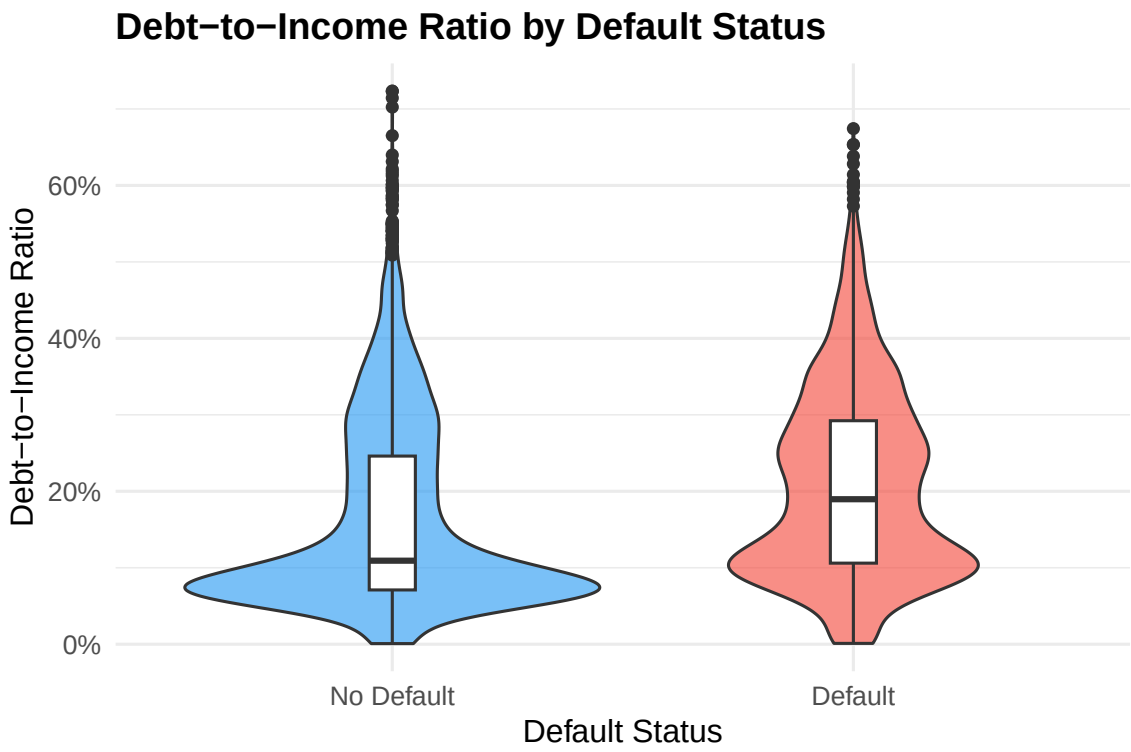

Figure 6: Figure 6: Debt-to-Income Ratio by Default Status

Section 9: Default Rate by Product Category

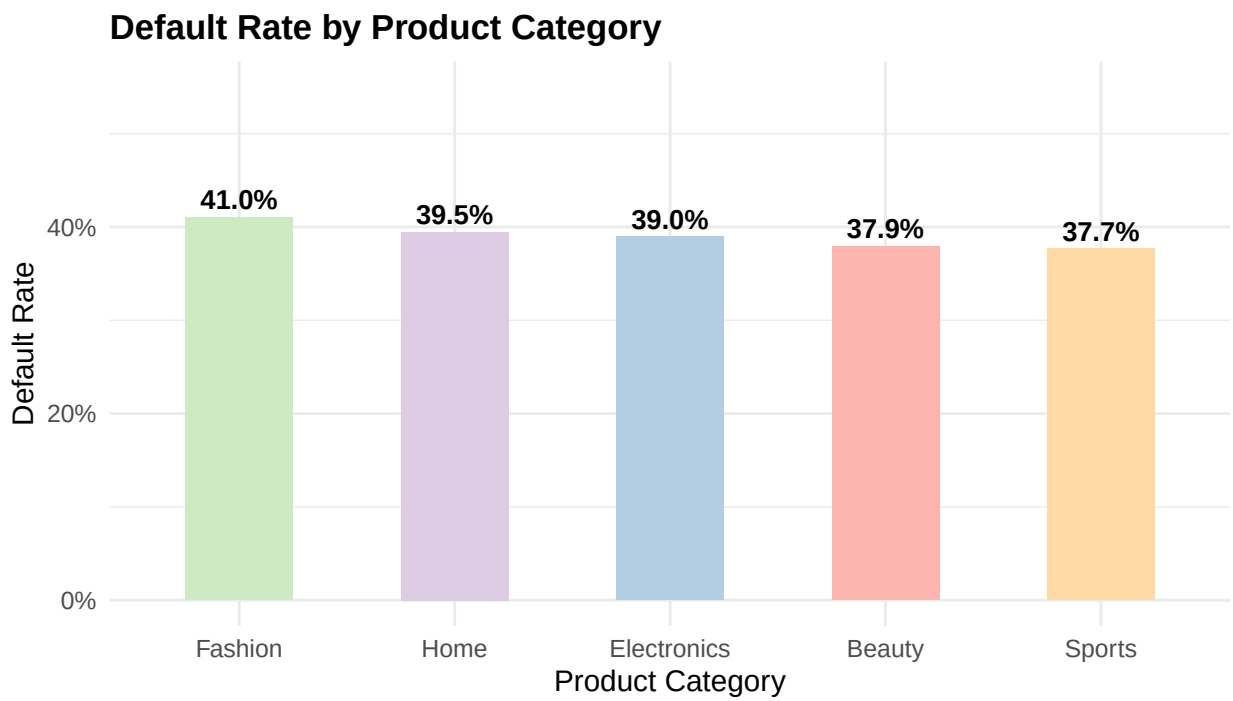

Figure 7: Figure 7: Default Rate by Product Category

Interpretation: Default rates across product categories are relatively similar, ranging from 37.7% (Sports) to 41.0% (Fashion). While Fashion purchases show a slightly higher default rate, the differences are not dramatic. This suggests that product category alone is a weak predictor of default — the nature of the purchase matters less than the financial profile of the buyer.

Section 10: Correlation Heatmap of Numerical Variables

Interpretation: The correlation matrix reveals several notable relationships. `risk_score` is positively correlated with `missed_payments` and `repayment_delay_days`, confirming that the risk score is constructed from behavioural repayment data. `monthly_income` and `debt_to_income_ratio` show a negative correlation — higher-income users tend to have lower debt burdens. Importantly, no two variables are extremely highly correlated (> 0.9), so multicollinearity is unlikely to be a major concern in modelling.

Section 11: Income Distribution by Default Status

Interpretation: Non-defaulters show a slightly higher income density at the upper end compared to defaulters. However, the distributions heavily overlap, indicating that income alone is not sufficient to distinguish defaulters from non-defaulters. This reinforces the need for a multivariate model that combines income with behavioural and demographic factors.

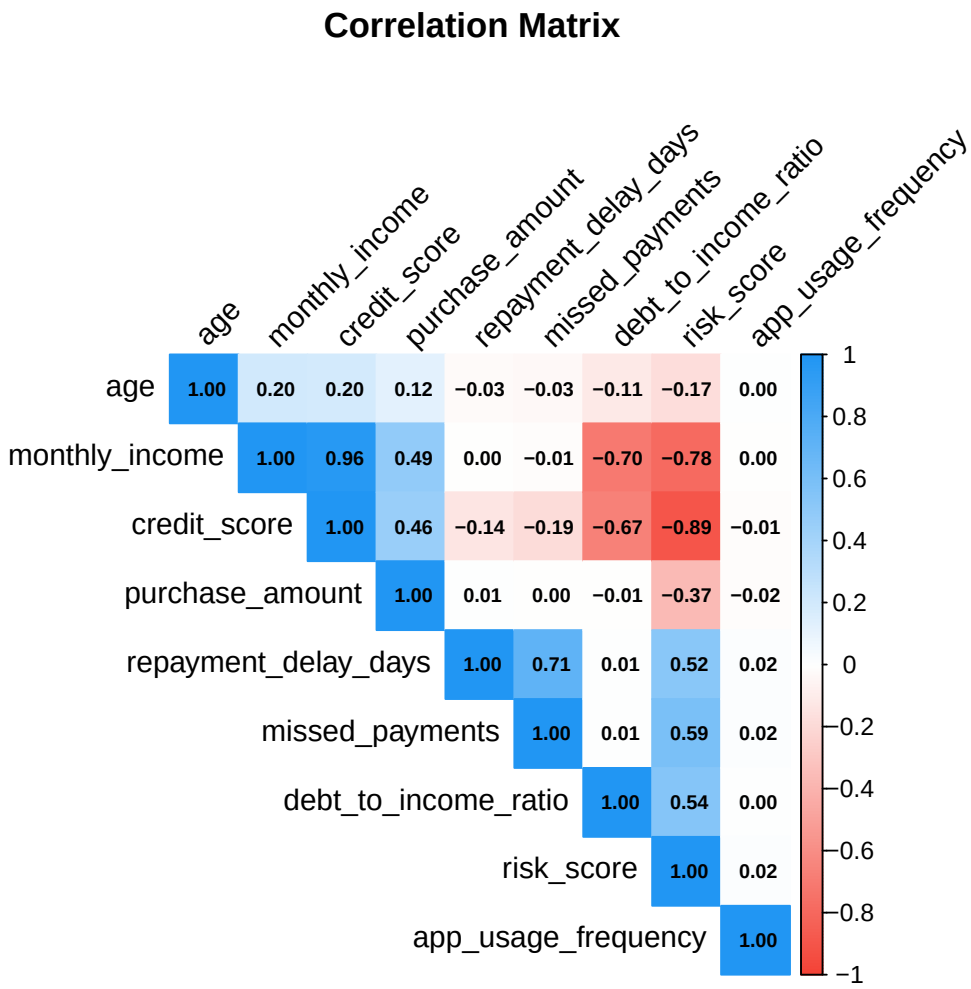

Figure 8: Figure 8: Correlation Matrix of Numerical Variables

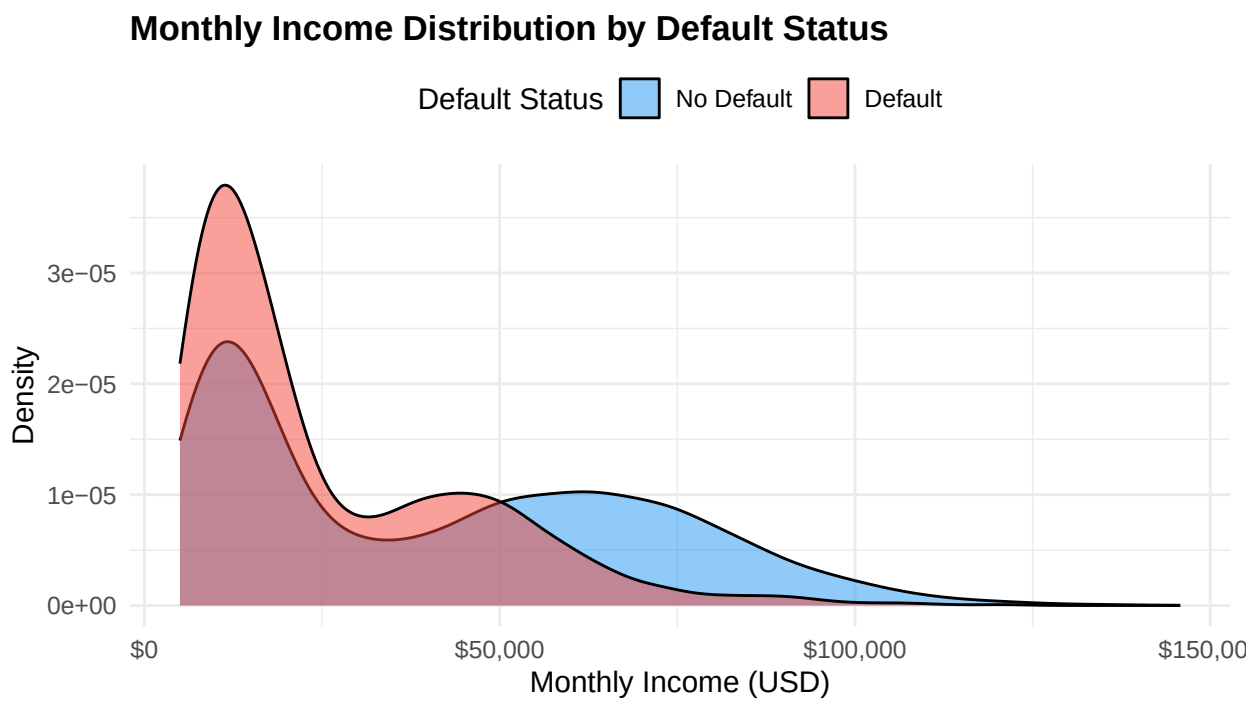

Figure 9: Figure 9: Monthly Income Distribution by Default Status

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Section 12: Monthly Transaction Volume Over Time

Interpretation: Transaction volumes are relatively stable across the observation period (2023–2024), with no dramatic seasonal spikes. The default rate (red line) also remains broadly consistent over time, hovering around 39%. This stability suggests the dataset captures a representative, steady-state snapshot of BNPL behaviour, making it well-suited for building a generalizable predictive model.

Summary of Key EDA Findings

Table 3: Table 3: Summary of Key EDA Findings

#	Key Finding
1	Overall default rate is 39.1% — moderate class imbalance requires handling during modelling
2	Unemployed and Student users have the highest default rates (~49%), nearly double that of Self-Employed users
3	Customer risk segment is highly predictive: Low Risk defaults at 4.5% vs. High Risk at 49.0%
4	Lower credit scores strongly associated with higher default probability

5	Missed payments and repayment delays are the strongest behavioural indicators of default
6	Higher debt-to-income ratio is a key financial stress indicator linked to default
7	Product category shows minimal variation in default rates — a weaker predictor

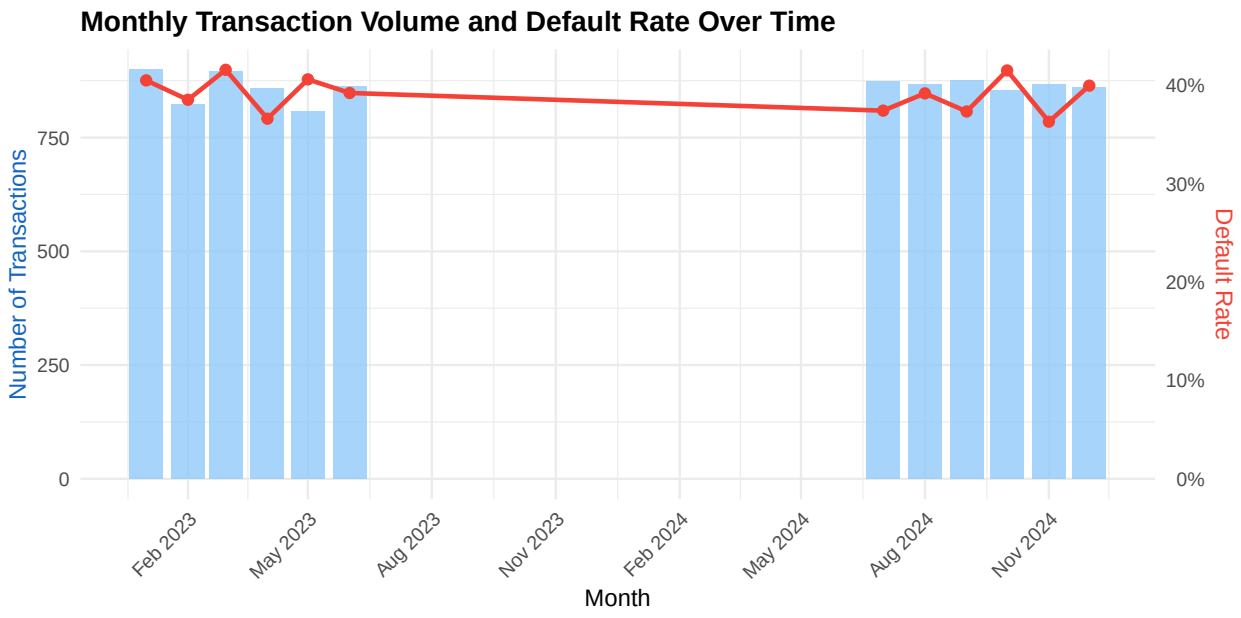

Figure 10: Figure 10: Monthly Transaction Volume and Default Rate Over Time